# Retail Orders: Queries by Hand

**Fall 26**

Calvin Williamson, Science and Math

The retail orders database has three tables. Each order is one customer buying one product. `customer_id` and `product_id` in `orders` point back to the `customers` and `products` tables.

Answer each question **by hand**, using only the tables below. Write your answer as a small table or a number. Later you will write a SQL query for each question and check your answers on the database.

## The tables

### customers

| customer_id | first_name | last_name |
|---|---|---|
| 1 | John | Doe |
| 2 | Jane | Smith |
| 3 | Robert | Johnson |
| 4 | Sarah | Williams |

### products

| product_id | name | price |
|---|---|---|
| 1 | Silk Blouse | 89.99 |
| 2 | Denim Jeans | 79.99 |
| 3 | Cashmere Sweater | 149.99 |
| 4 | Cotton T-Shirt | 29.99 |
| 5 | Wool Coat | 199.99 |

### orders

| order_id | customer_id | product_id | order_date |
|---|---|---|---|
| 1 | 1 | 3 | 2025-03-10 |
| 2 | 2 | 1 | 2025-03-12 |
| 3 | 3 | 4 | 2025-03-15 |
| 4 | 2 | 4 | 2025-03-18 |
| 5 | 1 | 2 | 2025-03-20 |
| 6 | 2 | 3 | 2025-03-25 |

## Level 1: one table

**1.** Which products cost less than $100? Give the name and price of each.

**2.** List all the products from most expensive to least expensive.

**3.** List the order number and date of every order Jane Smith placed.

**4.** How many orders were placed on or after March 15, 2025?

**5.** What are the average, highest, and lowest product prices?

## Level 2: more than one table

**6.** For every order, give the order number, the first name of the customer, and the name of the product.

**7.** Who bought the Cashmere Sweater? Give their first and last names.

**8.** What is the total amount of money from all the orders?

## Level 3: totals for each group

**9.** How much did each customer spend in total? List only customers who placed at least one order.

**10.** How many orders did each customer place? Include every customer, even one with no orders.

**11.** How many times was each product ordered? Include every product, even one that was never ordered.

## Think about it

**12.** Which customer has never placed an order? How did you find out?

**13.** Which product has never been ordered? How did you find out?

**14.** If you paired every customer with every product, how many pairs would there be? Explain how you got your answer.